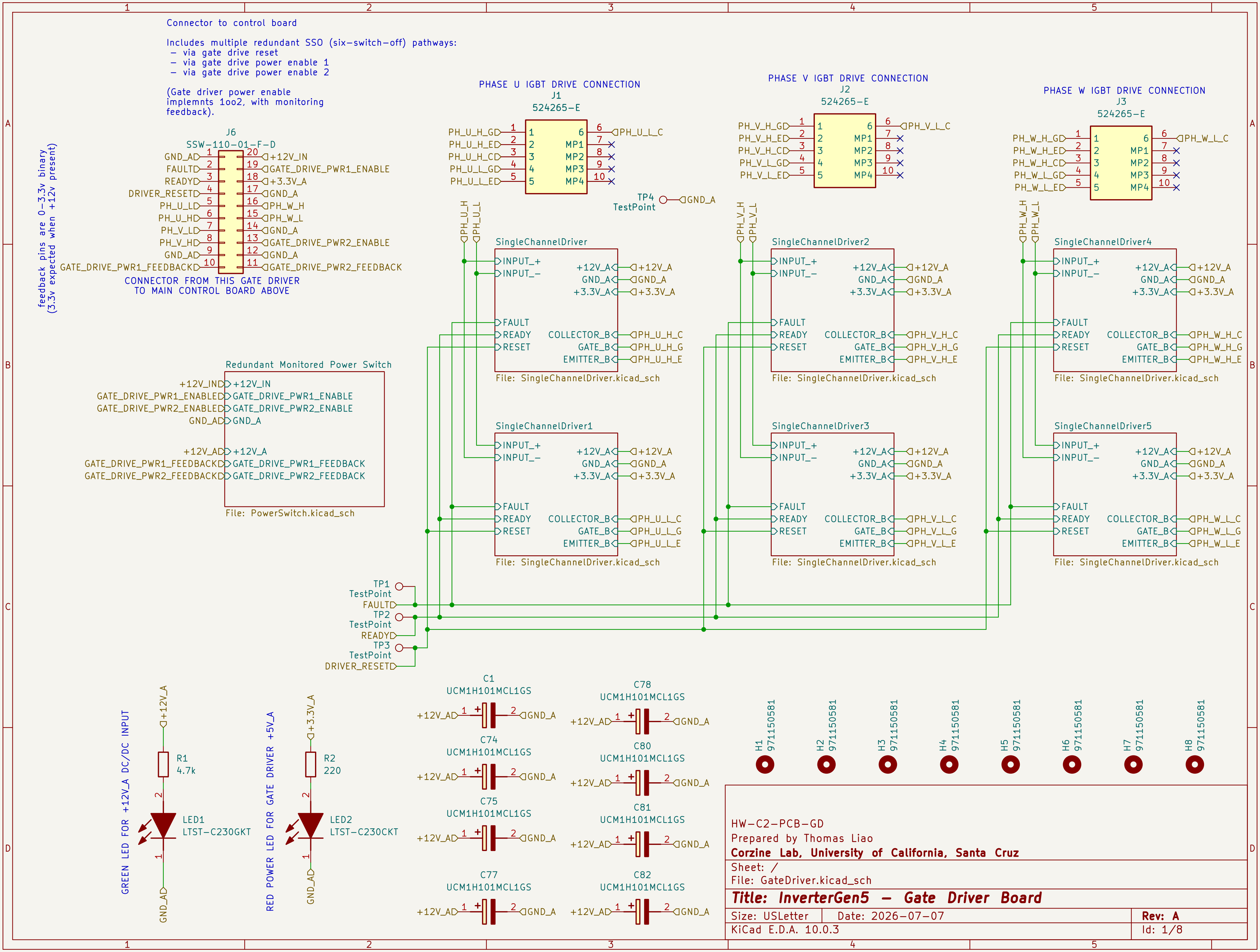
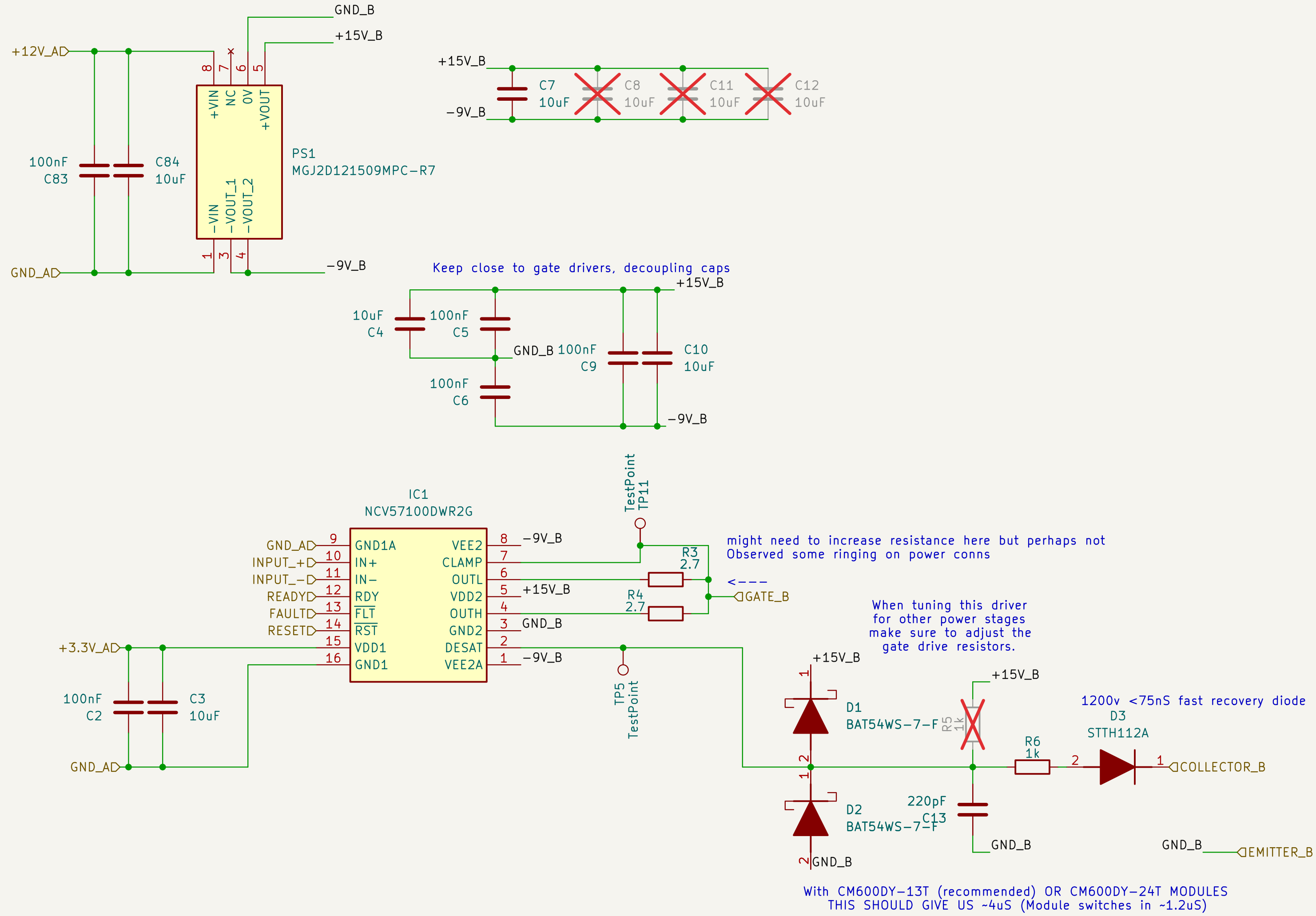

Connector to control board
Includes multiple redundant SSO (six-switch-off) pathways:
– via gate drive reset
– via gate drive power enable 1
– via gate drive power enable 2
(Gate driver power enable implemnts 1oo2, with monitoring feedback).
feedback pins are 0-3.3v binary (3.3v expected when +12v present)
J6
SSW-110-01-F-D
GND_A
FAULT
READY
DRIVER_RESET
PH_U_L
PH_U_H
PH_V_L
PH_V_H
GND_A
GATE_DRIVE_PWR1_FEEDBACK
+12V_IN
GATE_DRIVE_PWR1_ENABLE
+3.3V_A
GND_A
PH_W_H
PH_W_L
GND_A
GATE_DRIVE_PWR2_ENABLE
GND_A
GATE_DRIVE_PWR2_FEEDBACK
CONNECTOR FROM THIS GATE DRIVER TO MAIN CONTROL BOARD ABOVE
PHASE U IGBT DRIVE CONNECTION
J1
524265-E
PH_U_H_G
PH_U_H_E
PH_U_H_C
PH_U_L_G
PH_U_L_E
PH_U_L_C
MP1
MP2
MP3
MP4
PHASE V IGBT DRIVE CONNECTION
J2
524265-E
PH_V_H_G
PH_V_H_E
PH_V_H_C
PH_V_L_G
PH_V_L_E
PH_V_L_C
PHASE W IGBT DRIVE CONNECTION
J3
524265-E
PH_W_H_G
PH_W_H_E
PH_W_H_C
PH_W_L_G
PH_W_L_E
PH_W_L_C
TP4
TestPoint
GND_A
SingleChannelDriver
SingleChannelDriver1
SingleChannelDriver2
SingleChannelDriver3
SingleChannelDriver4
SingleChannelDriver5
INPUT_+
INPUT_–
FAULT
READY
RESET
+12V_A
GND_A
+3.3V_A
COLLECTOR_B
GATE_B
EMITTER_B
File: SingleChannelDriver.kicad_sch
Redundant Monitored Power Switch
+12V_IN
GATE_DRIVE_PWR1_ENABLE
GATE_DRIVE_PWR2_ENABLE
GND_A
+12V_A
GATE_DRIVE_PWR1_FEEDBACK
GATE_DRIVE_PWR2_FEEDBACK
File: PowerSwitch.kicad_sch
TP1
TestPoint
FAULT
TP2
TestPoint
READY
TP3
TestPoint
DRIVER_RESET
C1
C74
C75
C77
C78
C80
C81
C82
UCM1H101MCL1GS
GREEN LED FOR +12V_A DC/DC INPUT
R1
4.7k
LED1
LTST-C230GKT
RED POWER LED FOR GATE DRIVER +5V_A
R2
220
LED2
LTST-C230CKT
H1 971150581
H2 971150581
H3 971150581
H4 971150581
H5 971150581
H6 971150581
H7 971150581
H8 971150581
HW-C2-PCB-GD
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: GateDriver.kicad_sch
Title: InverterGen5 – Gate Driver Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

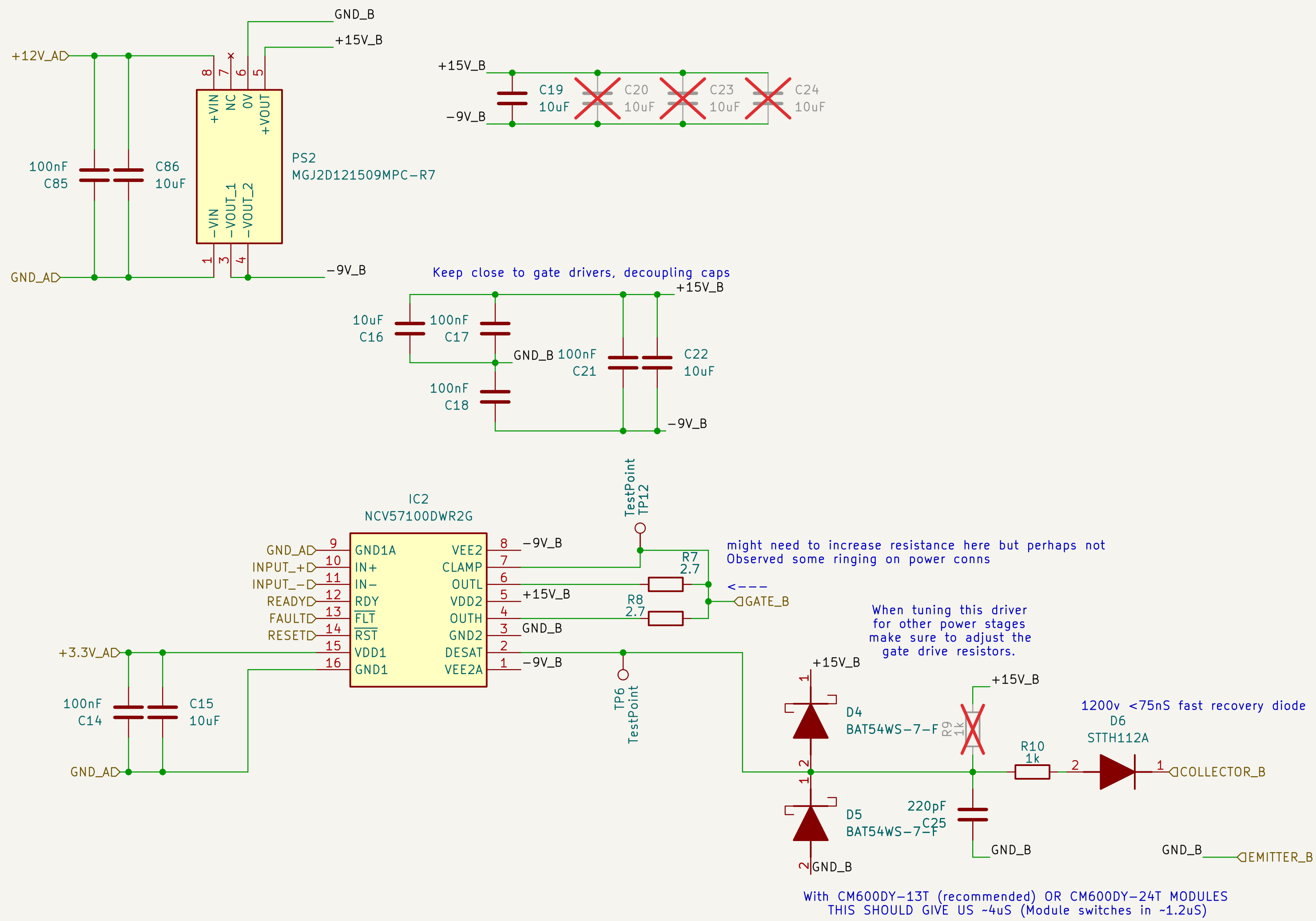
Sheet: /SingleChannelDriver1/

File: SingleChannelDriver.kicad_sch

Title: InverterGen5 - Gate Driver Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

KiCad E.D.A. 10.0.3 | Id: 2/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

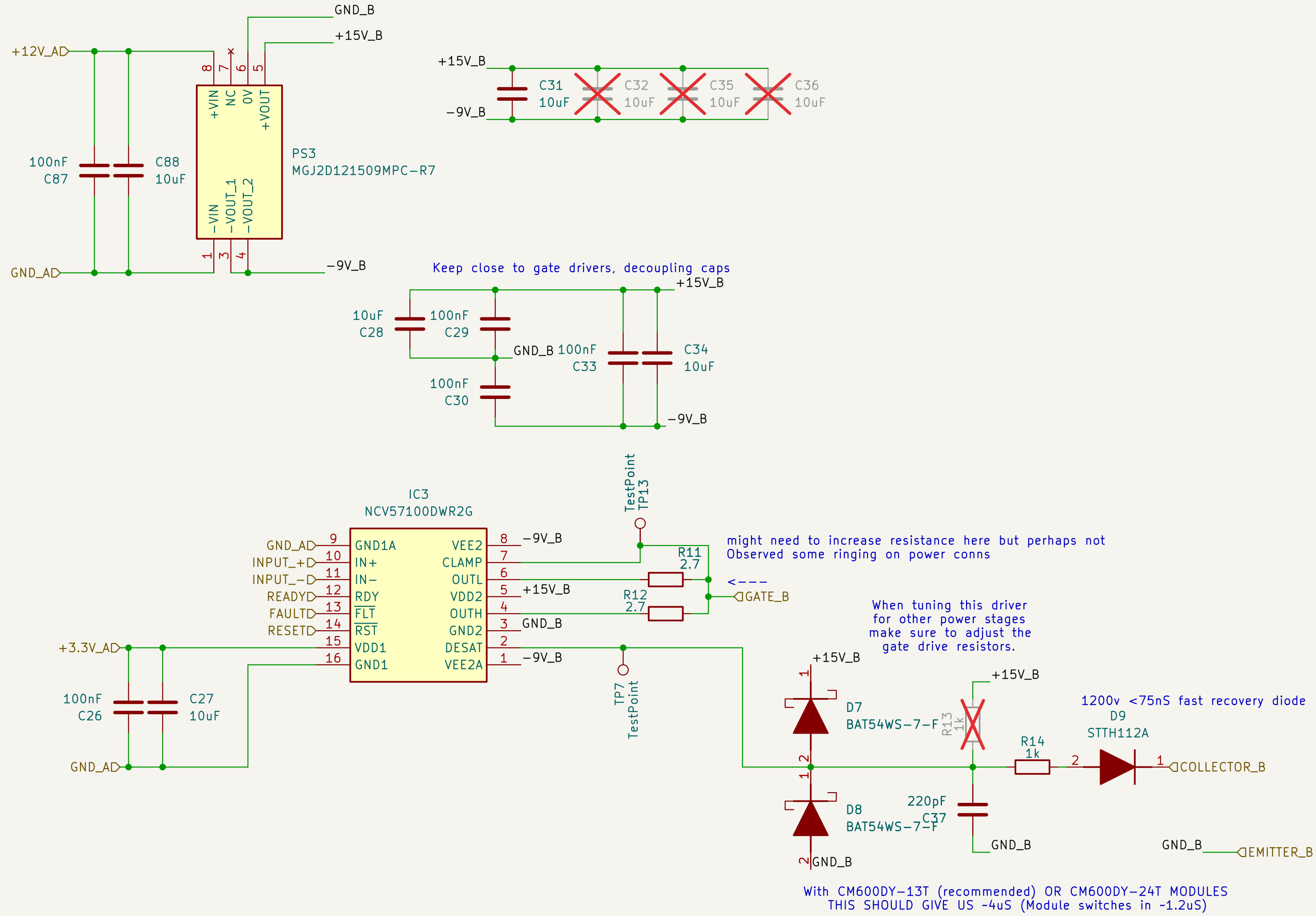
Sheet: /SingleChannelDriver2/

File: SingleChannelDriver.kicad_sch

Title: InverterGen5 – Gate Driver Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

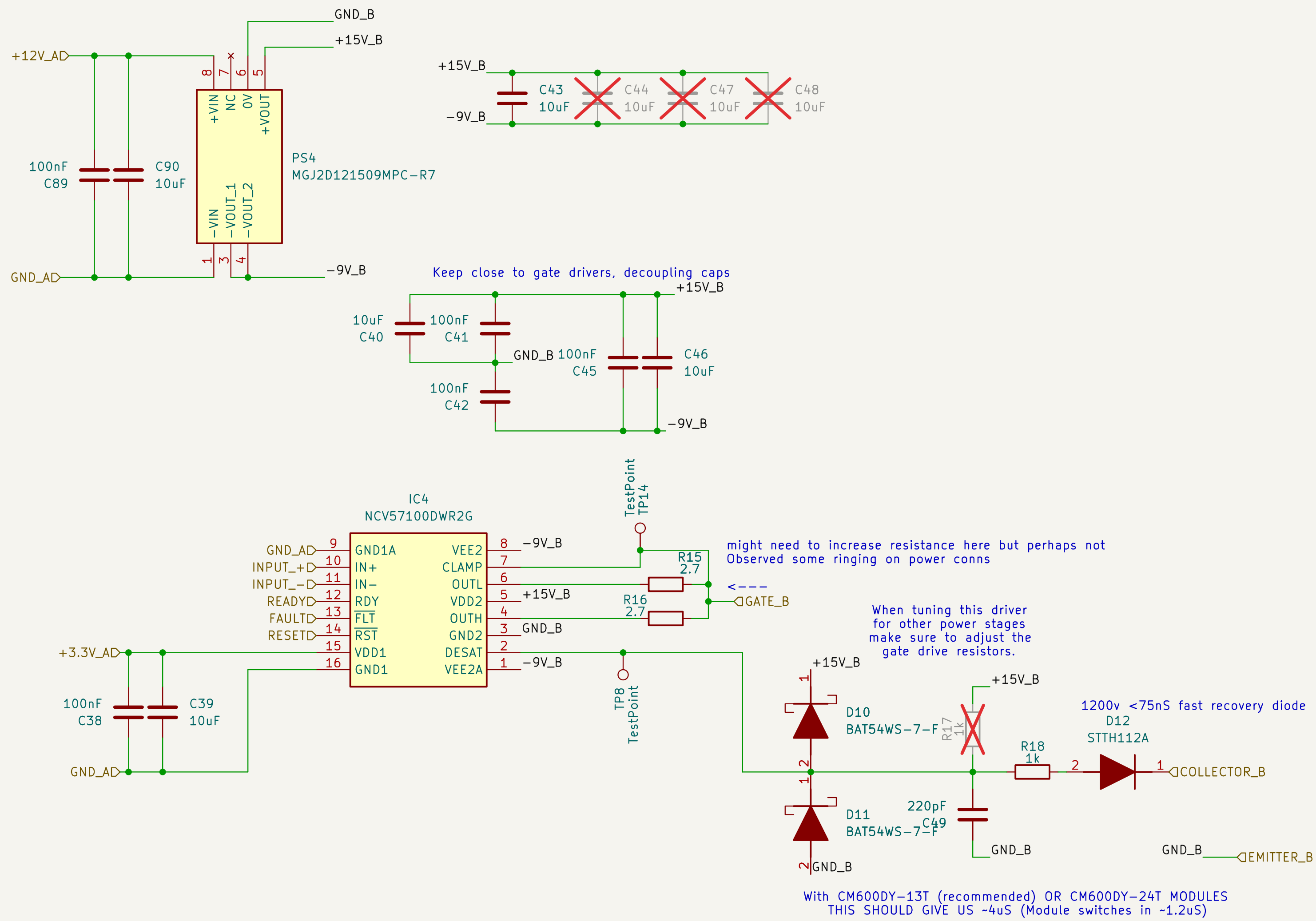
KiCad E.D.A. 10.0.3 | Id: 3/8



HW-C2-PCB-GD
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /SingleChannelDriver/
File: SingleChannelDriver.kicad_sch

Title: *InverterGen5 – Gate Driver Board*

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 4/8



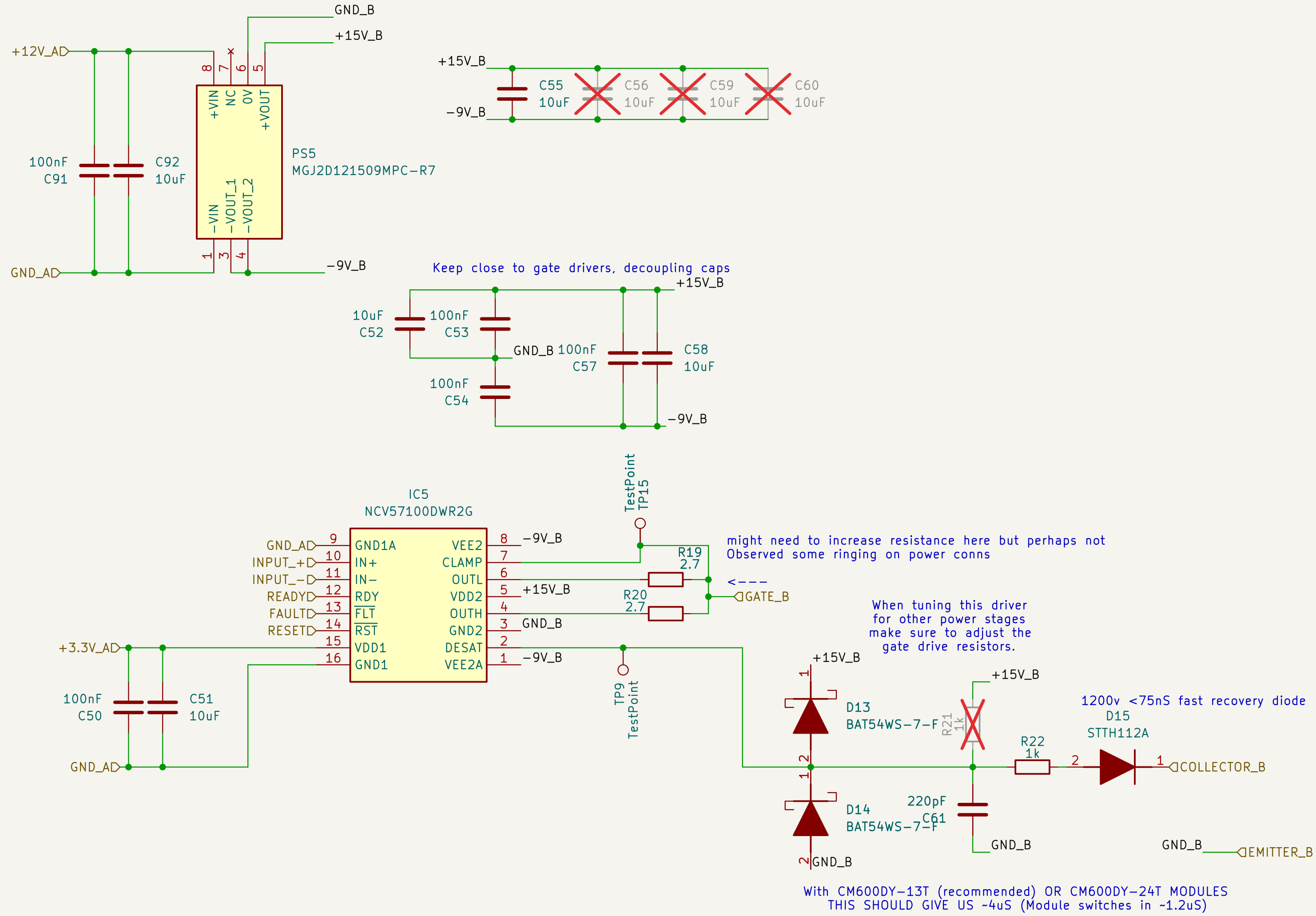
HW-C2-PCB-GD
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver3/
File: SingleChannelDriver.kicad_sch

Title: InverterGen5 – Gate Driver Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

KiCad E.D.A. 10.0.3 | Id: 5/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

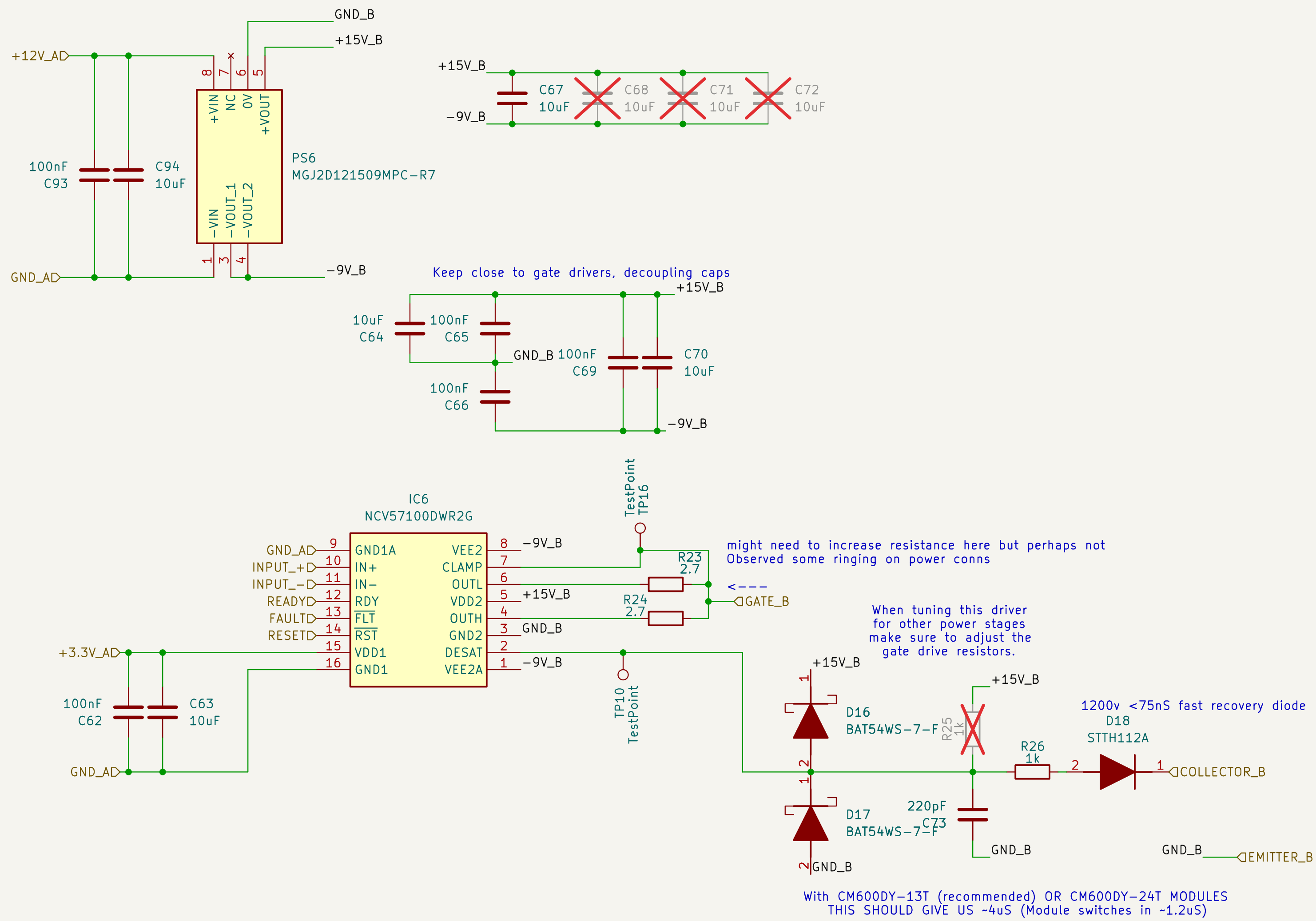
Sheet: /SingleChannelDriver4/

File: SingleChannelDriver.kicad_sch

Title: InverterGen5 - Gate Driver Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

KiCad E.D.A. 10.0.3 | Id: 6/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver5/

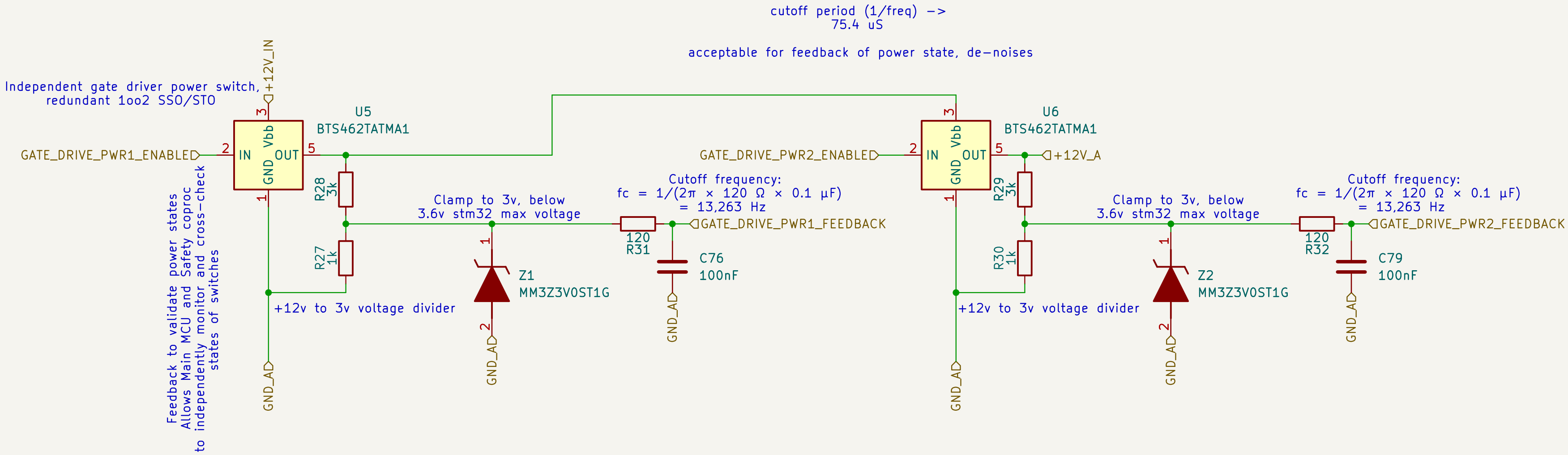
File: SingleChannelDriver.kicad_sch

Title: InverterGen5 - Gate Driver Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 7/8

This circuit implements a redundent pathway for both mcus to implement Six Switch Open / Safe Torque Off in addition to regular pwm break and gate driver reset.

The following switches are independent systems with feedback monitoring. Each MCU cross monitors returned inputs, and can validate if the switch failed to respond as commanded – thus removing latent failures from this safety subsystem.



HW–C2–PCB–GD
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Redundant Monitored Power Switch/
File: PowerSwitch.kicad_sch

Title: InverterGen5 – Gate Driver Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 8/8